

Lecture 1: Categorical Data

Categorical Data

We are dealing with multiple variables: a response Y and p predictors $X_1, \dots, X_p$.

- Y is a discrete random variable.
- Because Y is discrete, the standard linear model $Y = X\beta + \epsilon$ won't work, as it assumes a continuous response (and often normally distributed errors, which Y cannot be).

2 Types of Categorical Variables

1. **Ordinal:** The values (categories) in the support of Y with natural ordering.

- e.g., Letter grades in an exam: $A > B > C > D$
- e.g., Satisfaction survey (Likert scale):
"Very Good" > "Good" > "OK" > "Bad" > "Terrible"

2. **Nominal:** There is no logical ordering between the categories.

- e.g., Pincodes
- e.g., Voting preferences
- e.g., Mode of transport (car, bus, train)
- e.g., Color (red, green, blue)

In our models, the response Y will be categorical, and the predictors X can be a mix of continuous and categorical variables.

Binary Data (2 Categories)

This is the simplest case, often coded as 0-1 (e.g., Success/Failure, Heads/Tails).

- Let $P(\text{Success}) = p$, which is the parameter of interest.
- We have n independent trials: $Y_1, \dots, Y_n$ i.i.d. $\sim \text{Bernoulli}(p)$.
- Let $X = \sum_{i=1}^n Y_i$. Then $X \sim \text{Binomial}(n, p)$.

Parameter Estimation (MLE)

The Maximum Likelihood Estimator (MLE) for p is the sample proportion:

$$\hat{p}_{MLE} = \frac{X}{n} = \frac{\sum Y_i}{n}$$

- $E[\hat{p}_{MLE}] = p$ (The estimator is unbiased).
- $\text{var}(X) = np(1-p)$
- $\text{var}(\hat{p}_{MLE}) = \text{var}\left(\frac{X}{n}\right) = \frac{1}{n^2}\text{var}(X) = \frac{np(1-p)}{n^2} = \frac{p(1-p)}{n}$
- The (estimated) standard error is: $\text{s.e.}(\hat{p}_{MLE}) = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$

Asymptotic Properties

- **SLLN (Strong Law of Large Numbers):** $\hat{p} \xrightarrow{a.s.} p$
- **WLLN (Weak Law of Large Numbers):** $\hat{p} \xrightarrow{p} p$
- **CLT (Central Limit Theorem):** For i.i.d. X_i with $E[X_i] = \mu$ and $\text{var}(X_i) = \sigma^2$:

$$\frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \xrightarrow{d} N(0, 1)$$

- Applying CLT to our Bernoulli trials (where $\mu = p$ and $\sigma^2 = p(1-p)$):

$$\frac{\sqrt{n}(\hat{p} - p)}{\sqrt{p(1-p)}} \xrightarrow{d} N(0, 1)$$

Hypothesis Testing

We want to test $H_0 : p = p_0$ vs. $H_a : p \neq p_0$.

- **Test Statistic 1 (uses p_0):** Under H_0 , the variance is known ($p_0(1-p_0)$).

$$Z = \frac{\sqrt{n}(\hat{p} - p_0)}{\sqrt{p_0(1-p_0)}} \xrightarrow{d} N(0, 1) \quad \text{under } H_0$$

- **Test Statistic 2 (uses $\hat{p}$):** We can also estimate the variance using $\hat{p}$.

$$Z_{\text{est}} = \frac{\sqrt{n}(\hat{p} - p_0)}{\sqrt{\hat{p}(1-\hat{p})}}$$

By SLLN, $\hat{p} \xrightarrow{p} p_0$ (under H_0). By the Continuous Mapping Theorem, $\sqrt{\hat{p}(1-\hat{p})} \xrightarrow{p} \sqrt{p_0(1-p_0)}$.

By Slutsky's Theorem, if $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{p} c$ (a constant), then $\frac{X_n}{Y_n} \xrightarrow{d} \frac{X}{c}$. Here, $X_n = \frac{\sqrt{n}(\hat{p} - p_0)}{\sqrt{p_0(1-p_0)}}$

and $Y_n = \frac{\sqrt{\hat{p}(1-\hat{p})}}{\sqrt{p_0(1-p_0)}} \xrightarrow{p} 1$. Therefore, $Z_{\text{est}} = \frac{X_n}{Y_n} \xrightarrow{d} N(0, 1)$.

To get a $(1 - \alpha)100\%$ Confidence Interval

We use the second test statistic, as it doesn't depend on the (unknown) true p .

$$Z = \frac{\sqrt{n}(\hat{p} - p)}{\sqrt{\hat{p}(1 - \hat{p})}} \approx N(0, 1)$$

We construct the interval:

$$P\left(-z_{\alpha/2} \leq \frac{\sqrt{n}(\hat{p} - p)}{\sqrt{\hat{p}(1 - \hat{p})}} \leq z_{\alpha/2}\right) \approx 1 - \alpha$$

Rearranging for p :

$$P\left(\hat{p} - z_{\alpha/2} \frac{\sqrt{\hat{p}(1 - \hat{p})}}{\sqrt{n}} \leq p \leq \hat{p} + z_{\alpha/2} \frac{\sqrt{\hat{p}(1 - \hat{p})}}{\sqrt{n}}\right) \approx 1 - \alpha$$

The (approximate) $(1 - \alpha)100\%$ CI for p is:

$$\hat{p} \pm z_{\alpha/2} \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

Basically we reject if our estimator falls in one of the rejection regions(tails of the distribution)
We may need a randomized test to attain exact level test.

Multiple Categories (Multinomial Data)

Now, Y is a nominal variable with k possible values: $A_1, \dots, A_k$.

- Assume $Y_1, \dots, Y_n$ are i.i.d.
- Let $P(Y_i = A_j) = p_j$ for all $i = 1, \dots, n$.
- We have $p_j \geq 0$ for all j , and $\sum_{j=1}^k p_j = 1$.

Let X_j be the count of observations in category A_j :

$$X_j = \sum_{i=1}^n I(Y_i = A_j) = \#(Y_i = A_j)$$

The random vector $X = (X_1, \dots, X_k)^T$ follows a Multinomial distribution:

$$X \sim \text{Multinomial}(n, \mathbf{p}) \quad \text{where } \mathbf{p} = (p_1, \dots, p_k)^T$$

Note that $\sum_{j=1}^k X_j = n$.

The probability mass function (pmf) is:

$$P(X_1 = x_1, \dots, X_k = x_k) = \frac{n!}{x_1! x_2! \dots x_k!} p_1^{x_1} p_2^{x_2} \dots p_k^{x_k}$$

MLE for Multinomial Parameters

We want to find the $\hat{\mathbf{p}}_{MLE}$ that maximizes the log-likelihood. We have $k - 1$ free parameters, since $p_k = 1 - \sum_{j=1}^{k-1} p_j$.

$$\begin{aligned} l(\mathbf{p}) &= \log \left(\frac{n!}{x_1! \dots x_k!} \prod_{j=1}^k p_j^{x_j} \right) \\ &= C + \sum_{j=1}^k x_j \ln p_j \\ &= C + \sum_{j=1}^{k-1} x_j \ln p_j + x_k \ln p_k \\ &= C + \sum_{j=1}^{k-1} x_j \ln p_j + \left(n - \sum_{j=1}^{k-1} x_j \right) \ln \left(1 - \sum_{j=1}^{k-1} p_j \right) \end{aligned}$$

Taking the partial derivative w.r.t. p_j (for $j = 1, \dots, k - 1$):

$$\frac{\partial l}{\partial p_j} = \frac{x_j}{p_j} + \frac{x_k}{p_k}(-1) = \frac{x_j}{p_j} - \frac{x_k}{p_k}$$

Setting $\frac{\partial l}{\partial p_j} = 0$:

$$\frac{x_j}{p_j} = \frac{x_k}{p_k} \implies \frac{p_j}{p_k} = \frac{x_j}{x_k} \implies p_j = \frac{x_j}{x_k} p_k$$

This holds for all $j = 1, \dots, k - 1$. Now we use the constraint $\sum_{j=1}^k p_j = 1$:

$$\sum_{j=1}^k p_j = \left(\sum_{j=1}^k \frac{x_j}{x_k} p_k \right) = \frac{p_k}{x_k} \sum_{j=1}^k x_j = \frac{p_k}{x_k} (n) = 1$$

This gives $\frac{p_k}{x_k} = \frac{1}{n}$, so $\hat{p}_k = \frac{x_k}{n}$. Substituting back, $\hat{p}_j = \frac{x_j}{x_k} \hat{p}_k = \frac{x_j}{x_k} \frac{x_k}{n} = \frac{x_j}{n}$. The MLE for the parameter vector $\mathbf{p}$ is:

$$\hat{p}_{j,MLE} = \frac{X_j}{n} \quad \text{for } j = 1, \dots, k$$

Marginal Distributions

Consider the marginal distribution of a single count X_j from $X \sim \text{Multinomial}(n, \mathbf{p})$.

- **Result:** $X_j \sim \text{Binomial}(n, p_j)$.

- **Proof:**

- Let $V_i = I(Y_i = A_j)$ for $i = 1, \dots, n$.
- Each V_i is a Bernoulli trial:
- $P(V_i = 1) = P(Y_i = A_j) = p_j$
- $P(V_i = 0) = P(Y_i \neq A_j) = 1 - p_j$
- So, $V_i \sim \text{Bernoulli}(p_j)$.

- The total count is $X_j = \sum_{i=1}^n V_i$.
- Since the V_i are i.i.d., their sum is Binomial: $X_j \sim \text{Binomial}(n, p_j)$.
- This shows that the one-dimensional marginal of a multinomial distribution is a binomial distribution.

From this, we can easily find the variance of our estimator:

$$\hat{p}_{j,MLE} = \frac{X_j}{n}$$

$$\text{var}(\hat{p}_{j,MLE}) = \text{var}\left(\frac{X_j}{n}\right) = \frac{1}{n^2} \text{var}(X_j) = \frac{np_j(1-p_j)}{n^2} = \frac{p_j(1-p_j)}{n}$$